

Tree Height Triangulation Shootout

Measure a tree without climbing it.

Win condition: Closest to instructor reference height.

THE SCIENCE YOU ARE USING

Angles give height. You can find a tree's height without touching the top. Sight the treetop through a paper clinometer to read an angle, measure your distance to the trunk, and the geometry of a right triangle gives the height. This is remote sensing with a protractor.

Accuracy from method. Small sighting errors grow over distance, so careful technique matters: level your eye, sight steadily, measure distance honestly. The team closest to the instructor's reference height wins.

HOW TO BUILD AND RUN IT

- 01 Build the clinometer.** Fold a paper clinometer with a weighted string plumb line to read angles.
- 02 Use the marked distance.** Go to the spot the instructor pre-marked with the long tape and record that fixed standoff distance from the trunk.
- 03 Sight the treetop.** Look along the clinometer at the very top and read the angle steadily.
- 04 Compute the height.** tree height = eye height + (distance x tan(angle)). Measure your eye height with the 12 in ruler and use the marked standoff distance for the distance.
- 05 Check against reference.** Compare to the instructor reference. The closest estimate wins.

Safety: Outdoor boundaries. Allergy: Pollen sensitivity.

MATERIALS

Paper clinometers	4 (plus 2 spares)
String and washers	shared (1 roll string, 1 pack washers)
12 in rulers	4 (plus 2 spares; for eye height)
Long tape measure	1 (to pre-mark the standoff distance)
Clipboards	4 (plus 1 spare)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

Standoff distance from the trunk (marked by instructor): _____

Clinometer angle (average of several readings): _____

Eye height, and computed tree height = eye height + (distance x tan angle): _____

DEFEND YOUR ESTIMATE

What was your biggest source of error, and how close were you to the reference? _____

If you measured again, what is the ONE thing you would change to be more accurate? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	100